

An Ablation Study of Keypoint Contribution and Image Occlusion in the Keypoint-Promptable Re-Identification (KPR) Framework

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Abstract—The Keypoint Promptable Re-Identification (KPR) model was recently introduced, leveraging positive and negative keypoints during feature extraction in an ambiguous-target image, to address the multi-person ambiguity problem (MPA). While the original KPR paper conducts an ablation study to demonstrate the robustness of the model and its prompt optionality (full prompts compared to no prompts), this study will investigate the importance of each group of keypoints in the prompting process, providing partial keypoints to the model, and its robustness to synthetically occluded images along with missing keypoints. Using the Market-1501 dataset, and a pretrained KPR model with SOLIDER backbone with 93.25 Rank-1 accuracy and 96.59 mean average precision (mAP), we perform inference time ablations by masking specific keypoint groups (head, left-right arms, left-right legs, torso, arms, legs, top, non torso and arms, random, all). The results using full prompts serve as a baseline to identify which keypoint group carry the highest discriminative value, and expose the model’s weaknesses. We find that the torso is the most crucial body part for the model’s performance, as it contains important clothing cues. Finally, we propose a simpler inference method that compares to the baseline, by using a subset of keypoints consisting of : head, arms, and torso, partially mitigating the computational complexity required by the Re-ID task.

Index Terms—Computer Vision, Deep learning, Person Re-ID, Vision Transformer, Market-1501, Ablation Study

I. INTRODUCTION

Person Re-Identification (Re-ID) [1] is a fundamental and challenging task in computer vision, aiming to match a query image with other images of the same person in a gallery. Its complexity arises from images captured from different disjoint camera views, and many domains of application such as pedestrian flow analysis, sport video analysis, or video-surveillance. Despite the significant progress enabled by the use of CNNs and transformer-based models, the challenge persists in dense crowds with overlapping individuals where the target gets heavily occluded, or the presence of visually similar people, contributing to the multi-person ambiguity problem (MPA).

The recently introduced Keypoint Promptable Re-Identification (KPR) [2] framework proposes a new paradigm for addressing the MPA problem by including prompts in the feature extraction process in the form of positive and

negative keypoints to help better disambiguate the target individual. The paper was accompanied with the release of an annotated dataset, the Occluded PoseTrack-ReID, with a high proportion of occluded images, demonstrating the model’s full potential.

However, while the original KPR includes an ablation comparing the model when given full prompts and no prompts, it does not quantify the individual contribution of different keypoint groups. Subsets include (head, left-right arms, left-right legs, torso, arms, legs, top, non torso and arms, random, all), and understanding which of the latter carry the highest discriminative value is essential for designing a more efficient prompting strategy.

This work addresses these gaps through an ablation study using a pretrained KPR model with a SOLIDER backbone, with 93.25 Rank-1 accuracy and 96.59 mAP, evaluated on the Market-1501 [3] dataset that contains no occlusion. We conduct inference-time experiments by selectively masking keypoint groups in the prompts, and then synthetically occluding them on the image. Our objective is to determine which keypoint groups contribute most to the discriminative performance, exposing the limitations of the model; and to explore a simplified prompting strategy that maintains competitive performance.

II. RELATED WORK

A. Multi-Person Ambiguity and Non-Target Pedestrians

The challenge of distinguishing a target individual from other surrounding pedestrians has been explored in previous work attempting to solve the Non-Target Pedestrian (NTP) [4] problem. These approaches primarily reduce the influence of the distractor during the feature extraction process. However, such method don’t offer a mechanism to clearly identify which person in the image is the intended query target. On the other hand, KPR introduces target prompting, enabling the disambiguation of the target during both training and inference time.

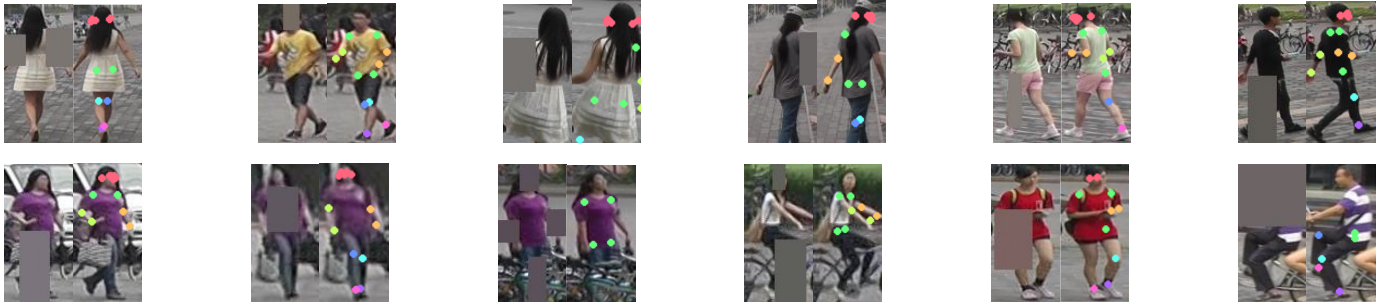


Fig. 1: Synthetically occluded images and their corresponding keypoint-suppressed prompts. The keypoint groups shown (left to right, top to bottom) are: arms, head, left arm, right arm, left leg, right leg, legs, torso, non-torso, non-torso and arms, random, and top.

B. Pose-Guided Person Re-Identification

Pose-Guided Person Re-ID introduces keypoints and part-based representations to improve the robustness under partial occlusion. Prior work like BPBreID [5] used pose labels only during training time, and cannot leverage them at test time. Others like HOREID [6] directly employ keypoints from a single person, but only focus on the exact keypoint location, failing to capture global appearance cues. While these methods incorporate keypoints, they don't offer a way to control the model's attention towards a target individual.

KPR improves on prior work by using the keypoints as a prompt, to both clarify the intended target with positive prompts, and distract the model away from non-target individuals with negative prompts. This combines the use of pose-guided Re-ID with prompting in vision in the context of transformer-based models.

III. DATA PREPARATION

Since the model does not attempt to match keypoints unless they are present in both images, the method of our ablation will be to preprocess keypoint prompts to remove keypoint groups corresponding to regions of the body, and render the contribution of that part of the network irrelevant, therefore ablating that model component. Additionally, we experiment with occluding pixels in a manner corresponding to the ablated prompts, so that we may compare these strategic occlusions to random occlusions. For the ablation study, it is only necessary to carry out this occlusion on the query set of images.

A. Strategic Occlusion

When we occlude a group of proximate keypoints, we generate a bounding box that contains all of them, and then find bounding boxes for non-occluded keypoints within that larger bounding box to exclude from occlusion. Bounding box edges are calculated on each axis as the midpoint line separating occluded and non-occluded keypoints; for keypoints closest to the image edge, their bounding boxes will extend to the image edge. In practice, the non-occluded regions within the larger occluding bounding box tend to be of negligible size due to their relative proximity on either axis to an occluding keypoint; however, without knowledge of the scale of each

keypoint, we would rather risk over-occluding than under-occluding, and the fact that some regions of the body share keypoints means that occluding one will always occlude some of any part that intersects it. For separated parts of the body like the arms or the head and legs, we occluded each in sequence to separate the occluded regions.

B. Random Occlusion

Randomly occluded images have a single occluding bounding box, whose area is a portion of the image's total area distributed according to $N(0, 25, 0.125)$ and clipped for overflows or underflows (this will occasionally result in no occlusion). Then, the aspect ratio of the bounding box is chosen as a ratio distribution of two variables normally distributed $N(1, 0.25)$ each clipped $[0.5, 1.5]$, for an average aspect ratio of 1 : 1 and a maximum of 3 : 1 at either axis. The pixel dimensions of the bounding box are then calculated, and any dimension exceeding a dimension of the image is clipped to it. The center of the bounding box is then chosen from a uniform distribution such that the bounding box does not overflow the edges of the image.

C. Keypoint Prompt Occlusion

To occlude a keypoint, we exclude it from the model's consideration by setting its confidence to zero. When strategically occluding with respect to keypoints, we will directly set each affected keypoint's confidence to zero, and when randomly occluding, we occlude whichever keypoints are located within the generated bounding box.

D. Image Occlusion

Whenever a pixel is within an occluding bounding box, we occlude the pixel by setting it to the mean pixel value of the image, such that the mean pixel value is unchanged, and all images later preprocessed with normalization with respect to their dataset or batch are affected by it equally.

IV. EVALUATION BENCHMARK AND METRICS

We evaluate our ablation on the widely used Market-1501 [3] benchmark dataset. We chose this dataset over others because of its absence of natural occlusion. This allows us to better control the synthetic occlusion, and which parts in the

prompt to omit. This would not be possible in other occluded datasets, such as the newly introduced Occluded PoseTrack-ReID where most images have significant invisible body parts.

The dataset is composed of three subsets : training set, testing set, and query set. The raw dataset contains 12,937 training images (751 IDs), 19,733 testing images (750 IDs), and 3,369 query images (750 IDs). To support KPR, the authors generated pseudo-labels in COCO [7] format using the PifPaf pose estimator [8]. However, we filtered the annotations from the gallery as some of them fail, and others are invalid. This filtering ensures all remaining annotations are valid, and offer reliable results for comparison.

A. Gallery construction and annotation filtering

The gallery set is derived from the test set and requires filtering to ensure consistent annotations. We apply the following filtering steps :

- **Invalid annotations:** Annotation files whose ID begin with -1 are discarded as they are considered "junk", meaning they have zero influence on the re-id accuracy. An image bbox detection is considered "good" if its Intersection over Union (IoU) with hand-drawn ground-truth bboxes is over 50%. Between 20% and 50% it is considered a "distractor". Below 20% is considered "junk".
- **Missing annotations:** If an image doesn't have a corresponding annotation file, or its content is empty, it is removed to maintain alignment in other images.

The same filtering method is used for both query and gallery sets. After processing, the final sets consist of :

- **Query:** 3328 images, 34 discarded due to missing annotations.
- **Gallery:** 14493 images, 16 discarded because of missing annotations, and 5275 invalid annotations.

B. Metrics

To measure the performance of the model, we report for each experiment the two standard retrieval metrics : The Cumulative Matching Characteristic (CMC) at Rank-1, and the mean Average Precision (mAP). Performance results are evaluated in a single-query setting.

V. EXPERIMENTS RESULTS DISCUSSION

A. Prompt optionality

Table I summarizes the impact of suppressing different groups of keypoints from the prompts and the corresponding KPR performance. The baseline which uses full prompts achieves a 93.68% Rank-1 accuracy and 84.32% mAP. Looking at all ablation configurations, we observe very small variations in both metrics, suggesting that the pretrained KPR model is strongly robust in the case of incomplete or suppressed keypoint information.

Interestingly, the torso suppression yields a very small drop in performance, and even scores the best Rank-1 accuracy among all groups, but also the lowest mAP. This suggests that

the torso provides the most stable spatial information to the model, and its removal disrupts the model's stability, resulting in easy examples dominating the Rank-1, and hard examples dominating the mAP. However, the performance decrease remains marginal, demonstrating the model's robustness.

As for the remaining keypoint groups, head suppression results in very small decrease in performance. Although it typically carries meaningful identity cues, the model seems to compensate with other keypoints using clothing and global appearance features. The removal of one or both legs, also yields negligible changes, with Rank-1 accuracy and mAP extremely close to the baseline. The same applies to the top (head, torso, and arms) suppressed.

Randomly suppressing the keypoint annotations doesn't impact the model. Indeed, even removing all the annotations yields a 93.44% Rank-1 and 83.84% mAP. This shows the prompt optionality of the model. However it is worth noting that the results are evaluated on the Market-1501 that contains no occlusion, and a single person per image. The absence of keypoint annotations does matter in occluded settings as shown in the ablation study in the original paper [2].

Overall, the results show that no single group is critical to maintain KPR's performance, and that the model shows strong robustness to missing prompts.

B. Image occlusion

Table II presents the performance of KPR model when keypoint groups are physically occluded within the query image. Gallery images remain unchanged, as the challenge is to match the occluded query image with its corresponding gallery identity. For consistency, we occlude the group in the image, and exclude the same keypoint group in the prompt. The model thus receives the keypoint prompts and visual information for all remaining keypoint groups. This challenges the model's robustness to visual degradation of the image. Therefore, the performance spread is significantly wider, unlike the prompt-suppression ablation that kept the visual information unchanged. The full unaltered images with full prompts achieve the same results as previously stated : 93.68% Rank-1 accuracy and 84.32% mAP.

Occluding the arms region leads to a drop in performance compared to the baseline with Rank-1 accuracy dropping from 93.68% to 81.68%, and mAP from 84.32% to 71.34%. This indicates that arms contribute to the model's ability to disambiguate individuals, because they contain clothing information in the sleeves.

The head, right arm, left arm, left leg, right leg, and legs are the least critical to the model's performance and have a performance comparable to the baseline in the range 86.64% - 93.68% Rank-1 accuracy and 76.39% - 84.3% mAP. When occluded individually, these group slightly impact the performance suggesting once again, that the model relies on clothing and upper body cues. All are within 7% difference from the baseline.

Suppressing the top (head, arms, torso) results in a major

TABLE I: Performance comparison of Keypoint-Promptable ReID with different keypoint groups suppressed.

Keypoints Group Suppressed	Rank-1 (%)	mAP (%)
Full prompts (baseline)	93.68	84.32
Arms suppressed	93.71	84.12
Head suppressed	93.56	84.26
Left arm suppressed	93.62	84.19
Right arm suppressed	93.71	84.24
Left leg suppressed	93.62	84.22
Right leg suppressed	93.68	84.29
Legs suppressed	93.68	84.30
Torso suppressed	93.74	84.06
Top suppressed	93.53	84.02
Non torso and arms suppressed	93.68	84.25
Randomly suppressed	93.59	84.24
All suppressed	93.44	83.84

TABLE II: Performance comparison of Keypoint-Promptable ReID with different keypoint groups occluded in images.

Keypoints Group Suppressed	Rank-1 (%)	mAP (%)
Full image (baseline)	93.68	84.32
Arms suppressed	81.68	71.34
Head suppressed	90.73	80.63
Left arm suppressed	89.47	79.40
Right arm suppressed	90.34	80.05
Left leg suppressed	93.62	84.22
Right leg suppressed	86.64	76.39
Legs suppressed	93.68	84.30
Torso suppressed	73.32	64.49
Top suppressed	14.77	9.93
Non Torso suppressed	59.78	52.87
Non Torso and Arms suppressed	69.95	61.68
Randomly suppressed	66.06	58.58
All suppressed	0.00	0.31

performance drop with 14.77% Rank-1 and 9.93% mAP, which is expected as it leaves no meaningful features for the model to discriminate between individuals. Only legs are left, and when suppressed, produce results identical to the baseline, meaning that they carry no discriminative value.

The torso occlusion also triggers a decrease in performance with 73.32% Rank-1 and 64.49% mAP, which contradicts the results from omitting it from the prompt that was the best achieving in the keypoint ablation. This suggests that while the keypoint prompt is optional, its presence in the image is crucial as it contains major clothing and pose cues that are useful to the model.

After showing that the torso is the most important body region, we evaluate how the model performs when given the torso only, and all other regions of the image occluded. The results are 59.78% Rank-1 and 52.87% mAP, which are relatively high, given that the model sees only the torso with its clothing cues. This aligns with our previous results showing the importance of the torso, with a significant drop when occluded. In addition, providing the arms along with the torso gives 10% in gains, when compared to the torso only. This validates again the reliance of the model on clothing cues.

Finally, randomly suppressing body parts using the method described earlier gives 66.06% Rank-1 and 58.58% mAP, which is low, but still above average. This is explained by the fact that the suppression of crucial body parts such as the torso is compensated by the suppression of other less relevant

body parts that don't impact performance like the legs, or the head. Suppressing the whole body from the image obviously results in 0.00% Rank-1 and 0.31% mAP, as the model can't see anything at all.

Overall, the most important region is the torso which causes the model performance to collapse when occluded. The head, arms and legs don't impact the model as it compensates with other visible regions. However, when all occluded at once (top suppressed), these cause the model to completely fail emphasizing the importance of the upper body. Randomly occluding images results in just above average performance, due to compensating body parts.

VI. ANALYZING THE CONTRIBUTION OF KEYPOINT GROUPS

Considering the results from both ablations, namely prompt suppression and image occlusion, we come up with the following ranking of keypoint groups from the least important to the most important to the model's performance.

- 1) Legs (left followed by right)
- 2) Arms (left followed by right)
- 3) Head
- 4) Torso

The ablation suggests that KPR does not require all 17 COCO keypoints annotations to achieve strong performance. By keeping only a subset of keypoints to reduce inference time, we can preserve the model's accuracy. Our proposed

keypoint subset consists of the torso, head, and arms, as they contain most of the clothing cues that help the model perform well. Omitting them represents the biggest drop in performance in both ablations. Thus, leg keypoints and their spatial information are mostly irrelevant to the model, and can be safely discarded, reducing the required number of keypoints from 17 to 13.

VII. CONCLUSION

In this work, we conducted a complementary ablation study to the original KPR paper. We evaluated the relative importance of individual keypoint groups under both prompt-suppression and image-occlusion conditions. Our findings show that while KPR is insensitive to incomplete or non-existent prompts, its performance significantly drops when the regions are physically occluded on the image. By combining the results from both ablations, we proposed a simpler inference strategy using only the torso, head, and arms subset of keypoints. This study provides an insight of KPR’s performance on extreme occlusion cases, contrasting with the Batch-wise Inter-Person Occlusion data augmentation (BIPO) that artificially generates inter-person occlusion, presented in the original paper.

VIII. LIMITATIONS AND FUTURE WORK

Although this study provides new insights into the contribution of prompts and keypoint groups for KPR, it was evaluated on a filtered subset of the original Market-1501 dataset. It would be interesting to explore the results on the enlarged version Market-1501 + 500K that contains an additional set of 500K distractors to make the retrieval more challenging. Furthermore, this dataset consists primarily of pedestrians in clean scenes at daytime. The KPR model might underperform when placed under more challenging datasets such as low resolution, poor lighting conditions, and cross-domain settings. The used pretrained weights are from SOLIDER which is the best performing for this dataset, however, it is worth investigating how the ImageNet backbone performs.

In addition, the synthetic occlusion introduces fixed shape and size rectangles, which doesn’t represent the real world as much as the BIPO would with artificial inter-person occlusion. Also, this study focuses on the major keypoint groups and some of their combinations. More combinations of keypoint groups are to be explored for a more thorough and detailed inference method. It is also possible to further narrow down the proposed keypoint subset by dynamically selecting keypoints based on an instance’s scene and context.

Since rebuilding the model’s code to omit parts classifiers was not within scope of our project, our ablation method does not materially impact the inference time of the model, as the part tokens are still calculated; they are simply ignored during the cosine similarity part matching due to low confidence. Therefore, we cannot currently measure the effect that excluding the legs has on the model’s inference speed.

Future work includes expanding the methodology to address all the limitations mentioned above, as well as fine-tuning

the model on the Market-1501 dataset both with and without the leg keypoints to test our hypothesis that the legs are unnecessary for re-identification.

ACKNOWLEDGMENT

This work was supervised and graded by Prof. Jochen Lang, who provided valuable feedback on the project proposal and presentation.

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